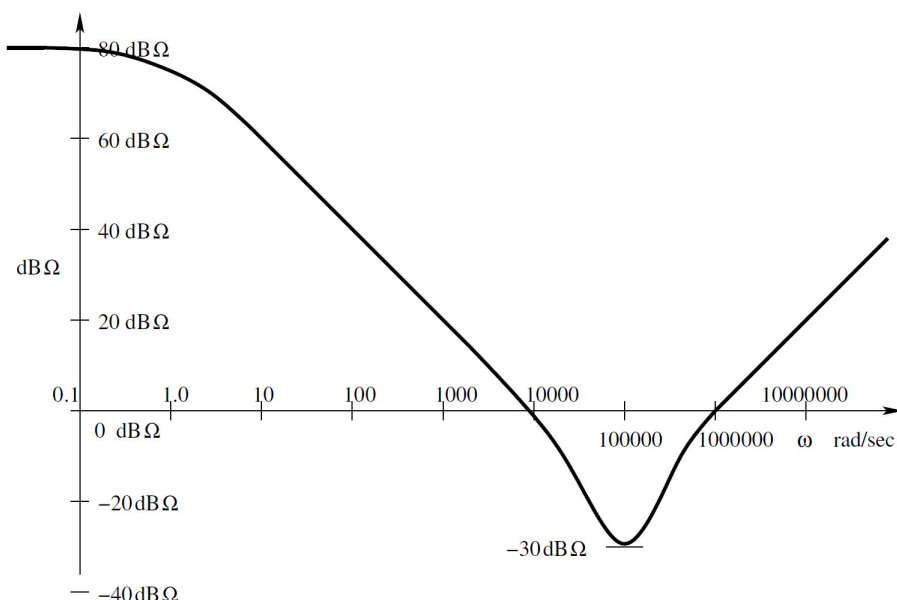
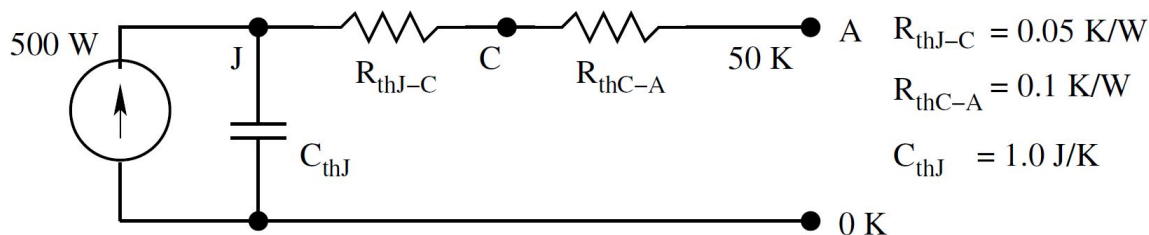


Assignment 4: Capacitors, Power Loss Computations and Thermal Design.

1. The figure shows the impedance of a capacitor as a function of frequency. (5)

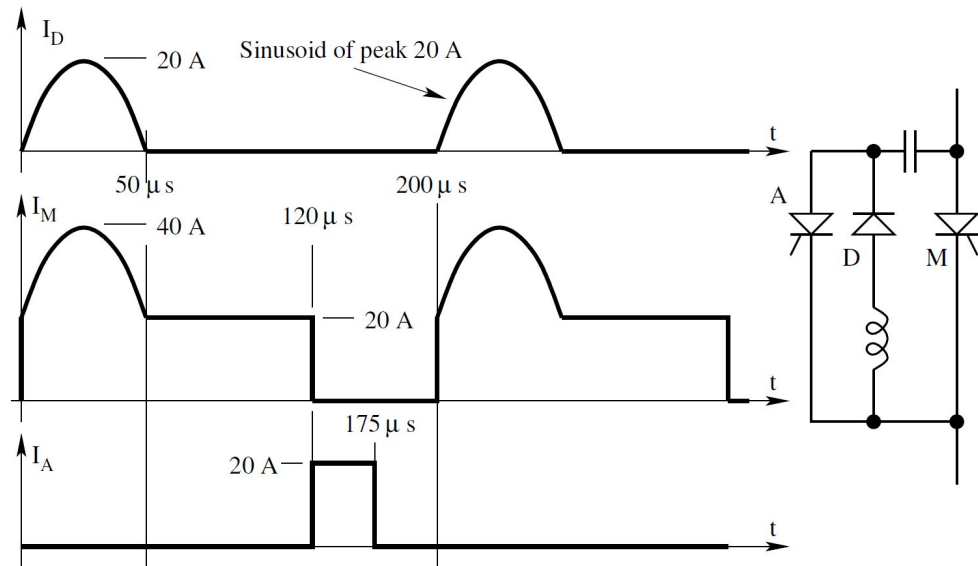


1. Draw the equivalent circuit of the capacitor including all the parasitic elements of the capacitor.
 2. Evaluate the values of all the parameters in the equivalent circuit.
2. The figure shows the thermal equivalent circuit of a semiconductor device with a steady state power dissipation of 500 W. The thermal resistances $R_{th,J-C}$, $R_{th,C-A}$ and $C_{th,J}$ are indicated. The ambient temperature is 50 °C. The maximum temperature the junction can withstand is 150 °C. (Treat $K = ^\circ C$). (5)

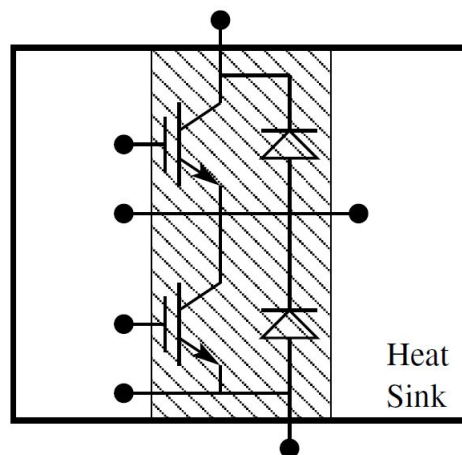


- A. Evaluate the steady state temperatures of the junction (J) and the case (C).
- B. Write the expression for the transient thermal impedance of the device, $Z_{th}(t)$. Evaluate $Z_{th}(t)$ at time $t = 0.1 \text{ s}$.

- C. Under the mentioned steady state condition, evaluate the excess transient power the device can dissipate for 0.1 s if the junction temperature is not to exceed 150°C .
3. The figure shows the current through the three power devices used in a chopper with commutation circuit. The power devices are M (SCR), A (SCR) and D (Diode). The SCRs may be modelled as a voltage drop of 1.5 V in series with a dynamic resistance of $20\text{ m}\Omega$. The diodes may be modelled as a voltage drop of 1 V in series with a dynamic resistance of $10\text{ m}\Omega$. (5)



- A. Evaluate the average and rms current through the devices M, A and D.
- B. Evaluate the average power loss in the devices M, A and D.
4. The figure shows a half bridge IGBT module mounted on a heatsink. It consists of two IGBT switches and two diodes. (5)



The module in a particular application dissipates 50 W of power in each of the diodes and 75 W in each of the IGBTs. The thermal resistances (junction to heatsink) are $0.15^{\circ}\text{C}/\text{W}$ for each of the IGBTs and $0.3^{\circ}\text{C}/\text{W}$ for each of the diodes. The thermal resistance of the heatsink to the ambient is $0.2^{\circ}\text{C}/\text{W}$. The ambient is at 30°C .

- A. Draw the thermal equivalent circuit of the device and heatsink in the ambient and mark the various thermal resistances.
- B. Evaluate the heatsink temperature and the junction temperature of each of the devices.